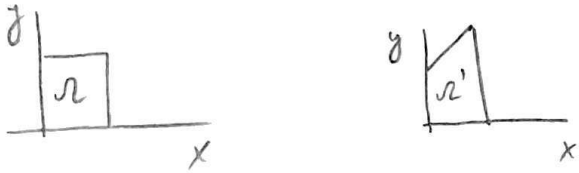
Prob. 2 Show that if $u(x,y)$ is a har-c f-n in Ω, then $u(\xi, \eta)$ is <u>not necessarily</u> harmonic in Ω', where Ω & Ω' are in the left half plane of $\mathbb{R}^2$ ($x>0$):

(1) $\xi = x^2$, $\eta = x+y$

(2) $\Omega' = \{ (x^2, x+y) : (x,y) \in \Omega \}$.

Hint: I got $\frac{\partial u}{\partial x} = u_\xi (2x) + u_\eta (1) = 2\sqrt{\xi}\, u_\xi + u_\eta$. A sample Ω & Ω' are

(3)



Sol-n:

First, by the hint for Pr 1-2, let me see 2 things:

(4) (i) $J = \begin{vmatrix} \xi_x & \xi_y \\ \eta_x & \eta_y \end{vmatrix} = \begin{vmatrix} 2x & 0 \\ 1 & 1 \end{vmatrix} = 2x \neq 0$ (except for $x=0$, $y \neq 0$, ok)

(ii) The Laplace eq-n $\Delta u = 0$, changes or not?:

(5) $u_{xx} = u_{\xi\xi}(\xi_x) + 2u_{\xi\eta}\, \xi_x \eta_x + u_{\eta\eta}(\eta_x)^2 + u_\xi \xi_{xx} + u_\eta \eta_{xx} =$ $\quad x = \sqrt{\xi}$

$= u_{\xi\xi} \cdot (2x)^2 + 2u_{\xi\eta}(2x)\cdot(1) + u_{\eta\eta}(1)^2 + u_\xi (2) + u_\eta (0) =$

$= 4\xi\, u_{\xi\xi} + 4\sqrt{\xi}\, u_{\xi\eta} + u_{\eta\eta} + 2u_\xi$, also, similarly,

(6) $u_{yy} = u_{\xi\xi}(0)^2 + 2u_{\xi\eta}(0)\cdot(1) + u_{\eta\eta}(1)^2 + u_\xi \cdot (0) + u_\eta (0) =$

$= u_{\eta\eta}$.

By (5) & (6) putting into $\Delta u = 0$, we have

$$\Delta u = (u_{xx}) + (u_{yy}) = (4\xi u_{\xi\xi} + 4\sqrt{\xi}\, u_{\xi\eta} + u_{\eta\eta} + 2u_{\xi}) + (u_{\eta\eta}) = 0 \Rightarrow$$

(7) $$\Delta u = 4\xi u_{\xi\xi} + 2u_{\eta\eta} + 4\sqrt{\xi}\, u_{\xi\eta} + 2u_{\xi} = 0$$ is NOT

a Laplace eq-n of $\nabla \cdot \nabla u(\xi, \eta) = 0$. Hence, the given transformation fails to preserve the harmonicity.

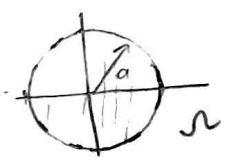
In a nutshell,

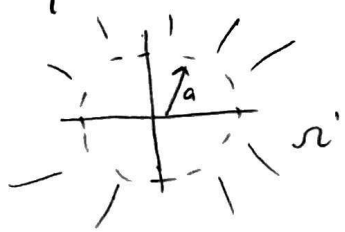
$u(x,y)$ is harmonic in $\Omega \not\Rightarrow u(\xi,\eta)$ is harmonic in Ω'

where Ω' is obtained by some transformation. $\blacksquare$

Prob. 4. Show that if $u(r,\theta)$ is a harm f-n in a circle, $r^2 < a^2$, then, $u(r', \theta')$ is harmonic in the <u>exterior</u> circle $r^2 > a^2$, where

(1) $$\begin{cases} r = a^2 \\ \eta = \theta \end{cases}$$





Sol.n: Let $u(r,\theta)$ be harmonic in $\Omega = \{r^2 < a^2\}$ i.e.

(1) $u(r,\theta) \in C^2(\Omega)$ & $\Omega = \{(r,\theta) : r^2 < a^2\}$

(2) $\Delta u = 0$

Then the used transformation:

(3) $$\xi = \frac{a^2}{r} \qquad \eta = \theta$$

Let's follow the hint in the introduction of this HW 10: